

Green Cloud: Supporting Sustainable Behavior by Helping Users Remove Unnecessary Photos from Cloud Storage Service

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ABSTRACT

Today's high-capacity data, especially digital photos, are readily generated and stored in cloud storage services. This excessive storage burdens data centers, which are major contributors to carbon emissions. Digital hoarding behavior — the excessive accumulation and reluctance to discard digital data — exacerbates this issue. Our study examines the impact of digital hoarding behavior on the task of deleting photos in cloud storage. The findings highlight the struggles users face and identify the need for better methods of deleting digital photos. We then focus on designing and evaluating features that assist users in photo deletion while making the experience satisfying. The goal of this study is to promote sustainable behavior in cloud storage services, where small actions, like deleting unnecessary photos, can lead to significant positive environmental impacts. Therefore, this study emphasizes individual behavior as a sustainable solution for addressing the environmental impact of dark data.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Dark data, Sustainability, Cloud storage service, Digital Hoarding behavior

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1 INTRODUCTION

Mobile digital technology developed rapidly due to the development of information and communication technology, which increased the amount of digital data produced by a myriad. In 2020, the total amount of data created and consumed worldwide is estimated to have reached over four-trillion gigabytes [1]. A significant part of this increase is attributed to digital photography, which has become an integral part of people's daily lives due to the widespread use of mobile devices [4]. People can easily take thousands of photos of their everyday lives [2]. As the amount of data produced increased, the need for large data storage also increased. In response, cloud storage services emerged, providing users with virtual space to store data on a secure physical server, which allows users to easily store data at a low cost [3]. Prior research shows that most cloud storage users sync data by default. This means large amounts of data are automatically stored on cloud storage which users consider as their primary data storage [5]. While this helps users store their digital data securely and readily, this has also increased dark data thrown into the cloud.

'Dark data' is digital data we put in the cloud and never access again [7]. Such data may increase energy consumption in data centers beyond necessity, resulting in an increase in carbon emissions [8]. Carbon emissions from global data centers have increased significantly over the years and are expected to account for 14% of the global carbon emissions in 2040 [9]. Environmental impact of global data centers is recognized as an important issue, and as a result, companies and researchers are finding ways to improve sustainable operation of data centers [6]. On the other hand, individual users' problem with dark data management in cloud storage is lacking in interest and solutions. Previous research shows that excessive storage of dark data affects how users manage digital data. Excessive data storage becomes a barrier for users when managing and deleting digital photos [18, 22]. It has also been established that designing a user experience which provides emotional satisfaction during the deletion process is essential [19]. Also considering that existing research indicates that the most common consumer use of cloud storage is storing photos [20], enhancing the photo deletion experience and promoting more frequent deletion could have a crucial impact on reducing dark data and supporting sustainability. Therefore, the study sought to analyze users' behavior on cloud

storage service and propose new cloud storage features that support energy-saving behavior by making it easier for users to delete unnecessary digital photos.

This study consists of two phases. In the first phase, we investigate motivations of digital hoarding and examine the influence of digital hoarding behavior. User survey was conducted to separate users into two groups, a group of users with hoarding tendencies and the other group of users without. This separation was used to compare responses from in-depth interview and gain insights into how both types of users currently perform the task of deleting photos. The second phase offers design features that support the task of deleting photos and evaluates their effectiveness.

2 RELATED WORKS: DIGITAL HOARDING BEHAVIOR

At the beginning of the study, we conducted a review of prior research addressing digital hoarding behavior. Hoarding disorder refers to the acquisition of, and failure to discard, possessions which appear to be useless or of limited value [10]. Motivations for hoarding behavior are Indecisiveness and Perfectionism. Indecisiveness makes it difficult for hoarders to decide what to keep and what to throw away. So, hoarders postpone the decision or make the ‘safe’ decision to keep the photos. That way, they avoid the mistake of throwing something away that they might later need. Perfectionism is related to worrying about the outcome of the decision when a mistake has been made [10]. Excessive possession is also closely related to emotional reasons as one can feel that their possession is an extension of the self [11] and that one can develop sentimental attachment [12].

Hoarding behavior can also be observed in the digital realm [13]. Existing research describes five motivations for digital hoarding: *keeping data for the future; keeping data as evidence; lazy and time constraints; emotional attachment to data; not my server - not my problem* [14]. Research states that people show emotional attachment particularly to photos, and so the task of deleting photos was perceived as distressing [14]. It is the memory that a photo holds which gives it personal meaning and relevance [2]. ‘Not my server - not my problem’ is a motivation that is distinct to the digital realm. Whereas physical hoarding can be put a curb when there is no more space [10], unlimited space of digital storage freed the concern regarding the issue [14]. ‘Lazy and time constraints’ motivation explained the general feeling of apathy towards deleting digital data. People are overwhelmed by the amount of data and felt that managing required too much effort [14]. Research also suggested the negative impact of digital hoarding on psychological well-being. Managing or deleting large volumes of digital data provoked stress and anxiety [14]. Thus, the need for design solutions that help the decision of users with digital hoarding behavior was apparent.

3 STUDY 1: USER RESEARCH

User research was conducted to identify the distinct challenges faced by users when deleting photos on cloud storage services, particularly users with digital hoarding behavior. In this section, we conduct a user survey and categorize users into two groups: those with stronger hoarding tendencies and those without. We

then recruit participants from each group for in-depth interviews to collect and compare responses from both groups of users.

3.1 User Survey

3.1.1 Method. We conducted a user survey to categorize users into two groups for subsequent in-depth interviews. Survey collected users’ hoarding tendency and satisfaction for cloud storage services. To assess respondents’ hoarding tendencies, we referenced a questionnaire from previous research on measurement of hoarding [15]. Specifically, we selected representative questions to reduce respondent burden, totaling five: two related to ‘Difficulty Discarding’, two addressing ‘Clutter’, and one pertaining to ‘Acquisition’. Regarding satisfaction, overall satisfaction with the service was measured according to the methods used in previous research on cloud storage service satisfaction [21].

3.1.2 Finding. The survey collected 45 completed responses, with hoarding tendency scores ranging from 1.4 to 3.8. A Shapiro-Wilk test was conducted to analyze the distribution, returning a statistic of 0.9566 and a p-value of 0.085. This suggests that the data did not significantly deviate from a normal distribution, indicating that using the mean for further comparative analysis is feasible. We divided the users into two groups based on the mean hoarding score of 2.49. Users with a hoarding score above 2.49 were classified as those with stronger hoarding tendencies (shortly termed ‘Hoarders’), resulting in 24 Hoarders and 22 Non-Hoarders. An independent T-test was conducted, revealing a significant difference in satisfaction scores between the two groups. Users with strong hoarding tendencies showed lower satisfaction scores ($M=3.38$, $SD=0.82$) compared to those without hoarding tendencies ($M=3.95$, $SD=0.95$); $t(44)=-2.21$, $p<.05$ (Table 1).

3.2 In-depth Interview

3.2.1 Method. In accordance with the two user groups, in-depth interviews were conducted to understand the challenges users face when deleting photos from cloud storage services. Two participants from each group were recruited for these in-depth interviews to compare differences between users with and without hoarding tendencies. The interviews followed a semi-structured format, covering topics such as the purpose of using photo cloud services, methods of managing storage capacity, types of stored photos, file organization and deletion methods, as well as satisfaction and evaluation of the currently used service.

3.2.2 Finding. When asked about the difficulties of deleting photos, participants’ responses aligned with results from previous research. The process of deleting photos was not worth the time and effort it takes. “*The number of photos exploded and managing them feels like a huge hassle.*” (P3, Hoarder). Also, participants showed they lack the motivation to delete and organize photos due to it being unapparent whether the storage space is clustered or not. “*Storage space seems organized. It’s hard for me to feel the sense clustered space.*” (P4, Hoarder). Emotional attachment was also visible on both user groups. However, hoarders showed far deeper emotional attachment, expressing reluctance to delete slightly different duplicates of photos taken in one session. “*I’ve had experiences where I hesitated to delete photos of my friends.*” (P1, Non-Hoarder), “*Even if*

Table 1: User Satisfaction with Cloud Storage Services Based on Hoarding Tendencies

Dependent Variable	Groups	Sample Size	Mean	Standard Deviation	t	p
Satisfaction on cloud storage use	Hoarders	24	3.38	0.82	-2.21*	0.02
	Non-Hoarders	22	3.95	0.95		

^a $p < .05$

photos look somewhat similar but have slight differences, I hesitate to delete any.” (P3, Hoarder). Emotional attachment seemed to emerge from memories that a photo provoked. “All these photos hold memories, and once I discard them, I feel like they’ll disappear from my memory.” (P4, Hoarder). Interestingly, some participants showed motivations being moderated when offered with acceptable reasons such as encouragement from close friends and family. “I hesitated to delete, but my family told me they were unnecessary, so I did.” (P4, Hoarder). Overall, three of the five motivations for digital hoarding, mentioned in previous research, were apparent when using cloud storage service [14].

Users’ go-to method of deleting photos from cloud storage service was through scrolling up and down and selecting what seems to be a redundant photo. This method was not ideal as it is hard to recognize the full content of a photo through grid-view. Additionally, swiping through each photo was used to reminisce old memories, during which deletion sometimes followed. Participants also showed priorities for selecting which photos to delete. We were able to categorize different groups of unnecessary photos. Overlapping photos and blurry photos were grouped as *useless photos*. Images of documents and screenshots were grouped as *single-use purpose photos*. *Old photos* were photos that had been taken a long time ago. Users with hoarding tendency saw *old photos* useless as the memory the photo once held had been forgotten.

4 STUDY 2: INTERFACE DESIGN AND EVALUATION

Second part of the study aimed to design and evaluate new cloud storage service features that support users to delete photos easily. In this section we turn the insights from study 1 into actionable questions, ideate design interface, build prototype and perform usability test to validate new design features for Green Cloud.

4.1 Interface Design

The insights from user research highlighted problems users faced when deleting photos. During the ideation phase, these insights were transformed into ‘How Might We’ (HMW) questions. HMW questions help in redefining problems into actionable queries. These questions were: “How might we make deleting process more engaging?”; “How might we help users decide whether a photo is valuable or not?”; “How might we provoke the need for cleanup before reaching storage limits?”

Based on our understanding of challenges users face and corresponding HMW questions, we laid out ideas for design solutions and developed two novel features for Green Cloud (Figure 1).

To make deleting process easier for users, we designed swipe-to-delete interface. Current experience of deleting photos is tiresome

to users. However, we observed that users enjoy going through their photos given the purpose was to reminisce memories. Swipe-to-delete feature combines that experience with deleting process. Using this feature, users are presented with their photos one at a time. Users can swipe up to delete the photo or swipe down to keep it. Swiping interaction was chosen as it can bring enjoyable experience for users, promoting engagement and continuous use overtime [16]. Also, the fluid interaction of swiping reduced the number of taps it takes to delete a photo, making it convenient for users.

In swipe-to-delete feature each photo is presented with brief information about the photo, helping users decide the value of the photo and mediate indecisiveness. Information includes date of when photo was taken, number of times it was downloaded from cloud and number of times it was viewed. According to findings from user research, the time that has passed since a photo was taken is one of the key pieces of information a user considers when deciding whether to delete a photo. Additionally, users perceive an item as useless if it has been abandoned and not used for a long period of time. Thus, the three pieces of information were presented to display the actual use and value of a photo. We suspected that the additional information might persuade users to easily delete photos with which users have comparatively weaker emotional bonds.

To instill a sense of need for cleanup before the cloud reaches its storage limits, we used visual representation of unorganized photos. Photos that haven’t been organized into a folder or shown on *swipe-to-delete* feature are categorized as unorganized photos. When viewed from grid view, these unorganized photos are presented slightly (by 10 or -10 degrees) tilted. Contrary to this, organized photos are presented upright. Our intention was to visually distinguish organized and unorganized photos and convey the clustered state of cloud storage space.

4.2 User Testing

4.2.1 Evaluation Methods. Usability test was conducted to verify the effectiveness and usability of the two features: *swipe-to-delete* and *tilted photos*. A total of nine users participated for usability test, 6 males and 3 females. These participants were recruited because they had hoarding tendency scores over 2.49 and were using cloud storage services to store photos. Based on the same criteria used in Study 1, all participants were categorized as users with hoarding tendencies. Test session was conducted in two stages (Figure 2). In the first stage, we examined the *swipe-to-delete* feature compared to two other cloud storage services and in the following second stage, we observed how users perceived the *tilted photos* feature.

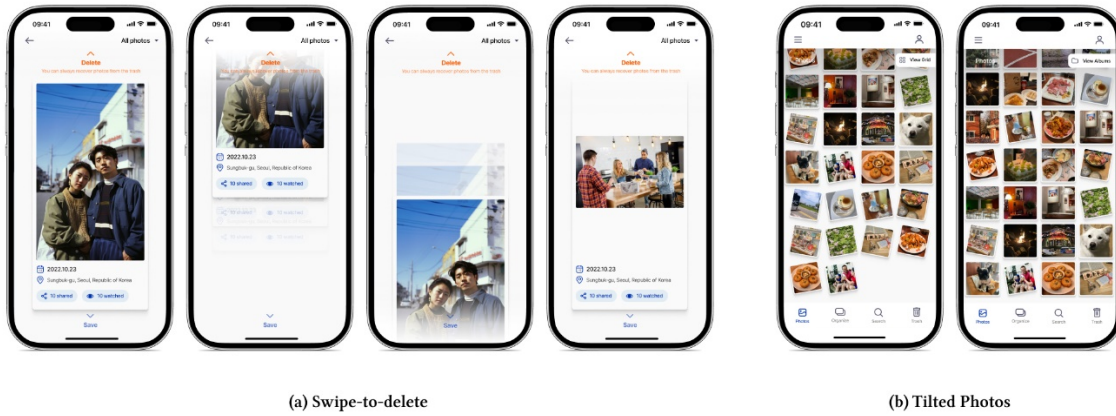


Figure 1: Interface designs of Swipe-to-delete and Tilted photos. ([https://youtu.be/gNvhGzGBFn0?si=\\$3l0TzrnM0jvH7XAt](https://youtu.be/gNvhGzGBFn0?si=$3l0TzrnM0jvH7XAt))

In the first stage, participants were instructed to perform photo deletion tasks. They were asked to delete photos that seemed useless with no time limit. For comparative analysis, participants performed the task on three different cloud services: Green Cloud’s *swipe-to-delete* feature, Google Drive, and Dropbox. Google Drive and Dropbox were selected based on previous research identifying them as the most used cloud storage services [5]. As the previous task can influence the subsequent one, we utilized counter balancing by dividing 9 participants into three groups, assigning different sequence of services for each group to vary the position of Green Cloud’s *swipe-to-delete* feature. Participants were given 3 sets of 50 pre-selected photos, one set for each service. These pre-selected photos were carefully chosen and evenly distributed considering the type of content and quality. Before performing each task, participants were thoroughly instructed about the interfaces and control to minimize the influence of familiarity. Number of photos deleted was collected to measure the effectiveness. Satisfaction with deleting method was evaluated using the After Scenario Questionnaire (ASQ) [17], complemented by the collection of qualitative data.

In the second stage, comparison between *tilted photos* and standard grid view was conducted to understand participants’ perceptions, aiming to confirm whether the intention behind the feature was clearly conveyed. Participants were given the *tilted photos* interface and asked about the impression and what participants thought the meaning of tilted photos was. Then they were given two screens, one with *tilted photos* and the other with standard grid view, followed by a question asking which one feels clustered.

4.2.2 Results. Table 2 illustrates the average number of photos deleted using each service. Green Cloud’s *swipe-to-delete* feature exhibited the highest number of photos deleted. Also, satisfaction scores from ASQ revealed that *swipe-to-delete* feature had highest average score of satisfaction in all three factors: satisfaction of

level of difficulty; satisfaction of time it took to completing task; satisfaction of information given (Table 3).

Qualitative data from participants showed that swiping interaction made deleting photos easier. “*Not needing to press multiple buttons to delete a photo was very convenient.*” (P2), “*Swiping seemed intuitive and easier than scrolling through hundreds of photos to remove photos*” (P5), “*Simple swiping method is very easy and seems less bothering when trying to delete photos*” (P9). Also, the *swipe-to-delete* feature turned laborious process of deleting photos into a more enjoyable experience. Participants even highlighted that they are willing to use the feature often. “*It’s more fun than the usual way. The ability to take a trip down memory lane and delete at the same time makes me think I’ll be checking it out more often. I enjoyed it although it took more time*” (P1), “*I prefer going through photos one by one when deleting, so this process of deleting with easy gesture was pleasing*” (P4), “*It’s great to be able to delete photos in a snap, the motion of flipping through is really satisfying. It’s fun*” (P5). Participants differed on which information about photo was useful or not. Some participants saw the date of the photo to be useful and some found the number of views to be more useful. Nonetheless, this implied that giving brief information along with the photo helps user decide the value of the photo. “*It was great to get information that doesn’t immediately come to mind just from looking at the photos. It made me think faster and really helped in deciding whether to delete a photo or not*” (P3). Although swiping interaction was easy and engaging, participants expressed the unfamiliarity of the interaction. Swiping down to proceed to next photo was perceived strange as horizontal swiping seemed familiar when moving to next photos. “*Having to swipe downwards to view the next photo feels a bit off since it’s different from usual habit. It gives a sense of doing something unusual*” (P7). Thus, it seems that for proceeding to the next photo, horizontal swiping should be adopted rather than swiping down.

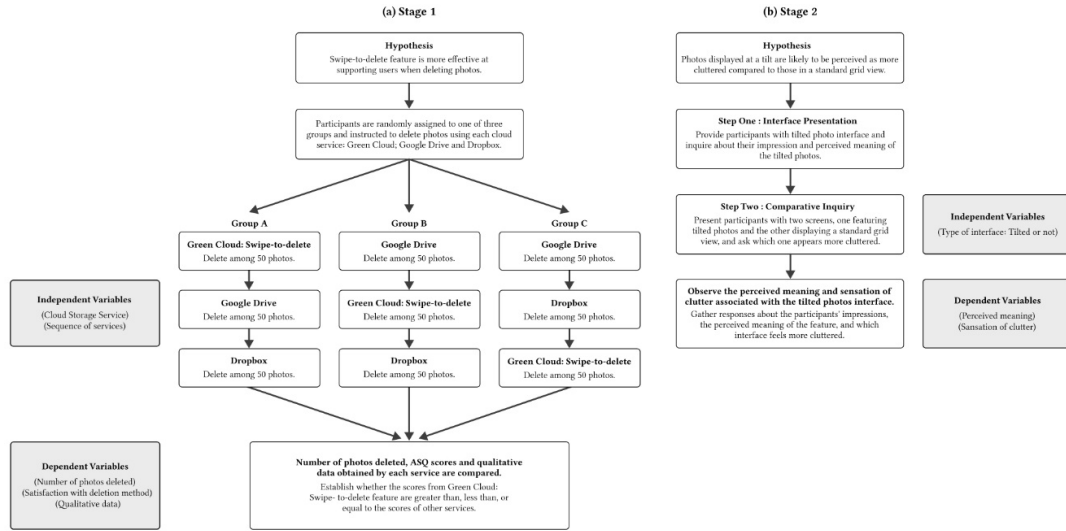


Figure 2: Experiment procedure diagram for usability test

Table 2: Descriptive Analysis of Number of Photos Deleted

	Swipe-to-delete Mean (SD)	Google Drive Mean (SD)	Dropbox Mean (SD)
Number of photos deleted	21.67 (3.97)	18.22 (7.64)	18.56 (12.42)

Table 3: Descriptive Analysis of Usability Score (ASQ)

Satisfaction of	Swipe-to-delete Mean (SD)	Google Drive Mean (SD)	Dropbox Mean (SD)
Level of difficulty	5.56 (1.01)	3.89 (2.03)	4.33 (2.12)
Time took for completing task	5.56 (1.74)	4.89 (1.90)	5.33 (1.94)
Information given	4.89 (1.54)	4.44 (1.74)	4.11 (1.96)

When asked about the *tilted photos*, participants successfully predicted the meaning of visual representation. Participants answered that tilted photos felt more clustered and needed to be organized. “It feels like a collection of poorly taken photos. Like a gathering of seldom-used pictures” (P8). However, some participants argued that the tilted photos were not distinct, suggesting that more apparent contrast may be necessary. “It’s hard to tell that the photo is tilted because both the photo and the background are white” (P3).

5 DISCUSSION

We investigated the impact of digital hoarding behavior on the photo deletion process in cloud storage services. Based on our findings, we introduced two features designed to ease the burden of deleting photos for digital hoarders and evaluated the efficacy of the features. In this section, we discuss insights from our study, its limitations, and contributions.

The first phase of the study focused on the impact of hoarding behavior on photo deletion in cloud storage services, leading to the development of Green Cloud with features designed for digital hoarders. Our research revealed specific challenges that hoarders

face when deleting photos, aligning with previous research on digital hoarding motivations. We determined that current cloud storage services do not adequately address the challenges associated with digital hoarding, highlighting the necessity for a deletion method tailored to the behavioral tendencies of digital hoarders.

In the second phase, we introduced two features—*swipe-to-delete* and *tilted photo*—and assessed their usability among hoarders. The *swipe-to-delete* feature transformed the onerous task of deleting photos into a more satisfying experience, possibly due to its fluid interaction and reduced effort, thus facilitating more deletion of photos. The long-term effectiveness of the *tilted photo* feature in promoting deletion remains to be determined.

Our study, limited by its small sample size, underscores the need for further research with a larger group of participants and an exploration of the long-term engagement and environmental impacts of these features.

As our study demonstrated, a tailored interface that mediates hoarding behavior has successfully nudged hoarders to delete more photos. The attainability of such behavior change, coupled with the limitations of our study, underscores the need for further research

and development in interface design. While existing research primarily addresses the issue of dark data from governance and technological perspectives, our study shows that a change in individual behaviors can also contribute to addressing the issue. Our study may contribute a unique perspective by emphasizing the role of personal behavior in mitigating excessive digital data accumulation.

6 CONCLUSION

Our study explored the relationship between digital hoarding behavior and cloud storage usage, leading to the development of novel features of Green Cloud service. Our research, through user surveys and interviews, identified the challenges of managing photos on cloud platforms for individuals with digital hoarding tendencies. Usability tests showed that Green Cloud's *swipe-to-delete* feature was more efficient and satisfying for users compared to traditional services. This feature was perceived as intuitive and engaging, helping users to more effectively manage and declutter their digital storage.

The results suggest that Green Cloud has the potential to encourage more sustainable digital storage practices. The positive responses from users indicate a move towards more mindful digital interactions. Green Cloud's design subtly guides users to organize their digital space, potentially signaling a shift towards sustainable digital habits. Ultimately, Green Cloud exemplifies how small, user-driven changes can make a significant contribution to a more organized and sustainable digital environment.

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REFERENCES

- [1] Branka Vuleta. 2021. How much data is created every day? +27 staggering stats. (December 2022). Retrieved from <https://seedscientific.com/how-much-data-is-created-every-day/>
- [2] Ine Mols, Elise van den Hoven, and Berry Eggen. 2014. Making memories: a cultural probe study into the remembering of everyday life. In Proceedings of the 8th Nordic Conference on Human-Computer Interaction: Fun, Fast, Foundational (NordiCHI '14). Association for Computing Machinery, New York, NY, USA, 256–265. <https://doi.org/10.1145/2639189.2639209>
- [3] Jiyi Wu, Lingdi Ping, Xiaoping Ge, Ya Wang, and Jianqing Fu. 2010. Cloud Storage as the Infrastructure of Cloud Computing. In Proceedings of the 2010 International Conference on Intelligent Computing and Cognitive Informatics (ICICCI '10). IEEE Computer Society, USA, 380–383. <https://doi.org/10.1109/ICICCI.2010.119>
- [4] Susan Enfield. 2022. How many photos will be taken in 2022? (March 2022). Retrieved from <https://news.mylio.com/how-many-photos-taken-in-2022/>
- [5] Nathan Sebastian. 2023. Usage & Trends of Personal Cloud Storage: GoodFirms Research. (October 2023). Retrieved from <https://www.goodfirms.co/resources/personal-cloud-storage-trends>
- [6] Hongyu Zhu and Dongdong Zhang and Hui Hwang Goh and Shuyao Wang and Tanveer Ahmad and Daijiafan Mao and Tianhao Liu and Haisen Zhao and Thomas Wu. 2023. Future data center energy-conservation and emission-reduction technologies in the context of smart and low-carbon city construction. Sustainable Cities and Society 89 (February 2023), 104322. <https://doi.org/10.1016/j.scs.2022.104322>
- [7] Tom Jackson, Ian R. Hodgkinson. 2022. What is 'dark data' and how is it adding to all of our carbon footprints? (February 2023). Retrieved from <https://www.weforum.org/agenda/2022/10/dark-data-is-killing-the-planet-we-need-digital-decarbonisation/>
- [8] Jones, N. 2018. How to stop data centres from gobbling up the world's electricity. Nature, 561, 7722 (2018), 163–166.
- [9] Charlotte Trueman. 2019. Why data centres are the new frontier in the fight against climate change. (Aug. 2019). Retrieved from <https://www.computerworld.com/article/3431148/why-data-centres-are-the-new-frontier-in-the-fight-against-climate-change.html>
- [10] Frost, R. O., & Gross, R. C. 1993. The hoarding of possessions. Behaviour Research and Therapy 31, 4 (May 1993), 367–381. [https://doi.org/10.1016/0005-7967\(93\)90094-B](https://doi.org/10.1016/0005-7967(93)90094-B)
- [11] Lita Furby. 1978. Possessions: Toward a theory of their meaning and function throughout the life cycle. <https://api.semanticscholar.org/CorpusID:142135894>
- [12] Randy O. Frost, Tamara L. Hartl, Rebecca Christian, and Nicole Williams. 1995. The value of possessions in compulsive hoarding: Patterns of use and attachment. Behaviour Research and Therapy 33, 8 (1995), 897–902. ISSN 0005-7967. [https://doi.org/10.1016/0005-7967\(95\)00043-W](https://doi.org/10.1016/0005-7967(95)00043-W)
- [13] Martine J. van Bennekom, Rianne M. Blom, Nienke Vulink, and Damiaan Denys. 2015. A case of digital hoarding. Case Reports (2015), bcr2015210814. <https://doi.org/10.1136/bcr-2015-210814>
- [14] George Sweeten, Elizabeth Silience, and Nick Neave. 2018. Digital hoarding behaviours: Underlying motivations and potential negative consequences. Computers in Human Behavior 85 (2018), 54–60. ISSN 0747-5632. <https://doi.org/10.1016/j.chb.2018.03.031>
- [15] Randy O. Frost, Gail Steketee, and Jessica Grisham. 2004. Measurement of compulsive hoarding: Saving inventory-revised. Behaviour Research and Therapy 42, 10 (2004), 1163–1182. ISSN 0005-7967. <https://doi.org/10.1016/j.brat.2003.07.006>
- [16] Xue Dou and S. Shyam Sundar. 2016. Power of the Swipe: Why Mobile Websites Should Add Horizontal Swiping to Tapping, Clicking, and Scrolling Interaction Techniques. International Journal of Human-Computer Interaction 32 (2016), DOI: 10.1080/10447318.2016.1147902
- [17] James R. Lewis. 1991. Psychometric evaluation of an after-scenario questionnaire for computer usability studies: the ASQ. SIGCHI Bull. 23, 1 (Jan. 1991), 78–81. <https://doi.org/10.1145/122672.122692>
- [18] Çağın Başkan. 2021. Entire history of you: Research and design to manage digital hoarding. M.S. thesis. Middle East Technical University.
- [19] Shin, Youngsoo, Lee Sunhwa, Kim Yeunju, Im Chaerin, Kim Jinwoo. 2015. A Study on User Experience(UX) in the Usage Context of Erasing. Korean Society for Emotion and Sensibility 18, 2 (June 2015), 65–74. <https://koreascience.kr/article/JAKO201524848982576.page>
- [20] Martin Armstrong. 2021. What's in the cloud? (September 2021). Retrieved from <https://www.statista.com/chart/25896/gcs-cloud-storage-services-usage/>
- [21] Kwangdoo Choi, Insu Cho, Heejun Park, Kiwon Lee, and Junmo Kang. 2013. An Empirical Study on the Influence Factors of the Mobile Cloud Storage Service Satisfaction. Journal of the Korean Society for Quality Management 41, 3 (Sept. 2013), 381–394. DOI: <https://doi.org/10.7469/JKSQM.2013.41.3.381>
- [22] Elizabeth Silience, Jordan A. Dawson, Richard D. Brown, Kerry McKellar, and Nick Neave. 2023. Digital hoarding and personal use digital data. Human-Computer Interaction 0, 0 (2023), 1–20. Taylor & Francis. DOI: <https://doi.org/10.1080/07370024.2023.2293001>